

Stress: The Family Heirloom You Shouldn't Pass To Your Kids

The aspiration of health across the human population is not subjective, but rather a persistent and objective goal that ignores social constructs – whether that be race, class, religion or gender. Early stages of human development provide a particular point of emphasis due to the recognised vulnerability of a fetus, baby, toddler or child (Richter et al., 2019). Hence, being able to provide the best environment to ensure a child's quality of development is one of the many reasons studies on the topic are crucial.

In this project, I will demonstrate a focus on stress as a risk factor that can act through both an internal and external mechanism contributing to the relationship between a parent and child, considering its projection on the child's development. Given the varied ways stress can be understood across contexts, this study will center on chronic stress, examined through the lens of emotion. Most emotions are universal, experienced by all individuals at some point in life, such as love, joy, fear, frustration, and grief. Yet only a few evoke a primal physiological response, and stress is among them. It is a sentiment that can turn toxic and if consistent and prolonged will physiologically deteriorate your body, spill out on your environment and by extension your development.

I was particularly drawn to the topic of stress because of its significant presence throughout my life. Like many of my peers, I associate it with exam season, however, it is also a marker in my cancer journey. Diagnosed with a lymphoma at the age of 15, I piled on traditional stressful events of GCSEs and relationships with hair loss and immunity-preserving isolation. This meant I quickly became familiar with the constant

blanket of stress that greeted me as I woke and accompanied me through chemo and homework assignments. Some of my doctors even claimed that stress induced my cancer. An alternative-medicine specialist went as far as to read my energy and claim my mother's stress brought about my diagnosis. She felt both furious, having already sacrificed so much to afford my treatment, and guilty, haunted by the question of what if. Needless to say, I have danced with stress in all its forms and intensities for as long as I can remember, and I know it infects everyone.

Emotional stress can come from many sources and appear in many different forms, with not all of them being a risk factor for childhood development (McEwen, 2019). In fact, it can be seen as a useful tool when it comes to learning (Beck et al., 2025). This instance, termed eustress, provides a positive experience that stimulates enjoyment, boosts cardiovascular health and can sharpen cognitive function (Chu et al., 2024). Nevertheless, my investigation will consider stress from job-loss as the external aspect that considers the outermost sectors of the Bronfenbrenner model, and pregnancy as an internal source. Whilst conducting my research, I tried to focus solely on the mother as a source of stress in order to ensure a more controlled outcome alongside relating to my anecdote. However, the literature centred around mothers didn't feel sufficient, hence why I grouped the external stressor under paternal conditions rather than independently maternal.

In biology, stress is considered as "any physical or psychological stimuli that disrupt[s] homeostasis" (Chu et al., 2024). It is an umbrella term that can account for emotional factors, pollution's effect on the body, medical intervention by surgery or the exposure to traumatic events. All of which have a role on the normal processes in your

body, and respective to its duration, stress can damage your health and development. Characterized by the speed of the body's response, the axis responsible for a more immediate response is allied to the sympathetic-adreno-medullar system: SAM, which causes behavioural changes that are in tune with a fight or flight response (Chu et al., 2024). Explaining a greater focus, prominent attention, enhanced arousal and analgesia as your body's lungs expand and vasculature constricts (Chu et al., 2024).

When stress stimulates a slower response the hypothalamic-pituitary-adrenal axis takes over (HPA). The hypothalamus, an area of the brain, is stimulated to release a hormone called CRH (corticotropin-releasing hormone) which cascades into other systems that further release hormones (Chu et al., 2024). All these functions are essential to the body, however it is the persistence which induces chronic stress that becomes harmful. Chronic stress is complex, and acts on multiple pathways which dysregulates the crucial communication the brain has with the body, Mariotti (2015) states that this prolonged occurrence of stress alters the efficacy of the HPA axis damaging overall health. This constant exposure to these stress-related hormones and chemicals leads to resistance at the receptor level and this addition of physiological disruption can contribute to cardiovascular events, metabolic diseases alongside mood and mental disorders (Mariotti, 2015).

According to countless researches, it is evident that prenatal stress can produce devastational consequences that span generations (Coussons-Read, 2013). In order to appropriately discuss stress and pregnancy, it is important to recognise that stress can produce the outcome of miscarriage and other instances that are considered "pregnancy failure" (Parker & Douglas, 2010). Meta-analysis and reviews of research have

concluded that psychological stress is harmful to women in early pregnancy, and can result in spontaneous pregnancy loss (Qu et al., 2017). Nevertheless, if the pregnancy persists, we can observe the physiological response that stress promotes and have a long term effect on offspring health. The outcome of which can vary with the severity of the stress; abuse, life changes, or general daily hassles, yet seem to consistently impact the trajectory of the pregnancy and fetal development.

Specifically, prenatal stress has been associated with poor cognitive outcomes in offspring measured by tasks focusing on “working memory, spelling, reading (speed, accuracy, comprehension), response inhibition, selective attention, attentional control, social cognition, and verbal ability” (Roberts & Flouris, 2025). When looking at the timing of stressor exposure along the course of pregnancy, Francesconi, Roberts, and Flouri’s findings demonstrate that the exposure to the stressor, in whatever form, is more detrimental if introduced at a later time of gestation (Roberts & Flouris, 2025). It provides particular emphasis on reading, spelling and social communication of the child being attained by prenatal stress.

While the mother serves as a direct link within the innermost layer of Bronfenbrenner’s ecological systems theory, the stress itself seeps in from a more distanced environment to the child. Hence, prenatal stress itself is a fluid risk factor and can be designated based on its context. A more definite risk factor that resides in the mesosystem, is the influence of parental employment status, specifically the loss of a job. Job displacement produces a variety of strains whether that be an increase in family income, increased tension, stress or a shift in parenting. Therefore it is not surprising that there are consequences on early child development. An Irish study demonstrated

that the loss of a job from a parent “exacerbates problem behavior at ages 3 and 5” (Mari & Keizer, 2021). The study highlights that a loss of employment is a double edged sword that targets negatively parental behaviour on children from stress as well as less of an ability to invest in the child (Mari & Keizer, 2021). Nevertheless, the research is concluded with potential policy implications that focus on reducing the financial burden, focusing on public support for childcare and the buffer that subsidies could provide (Mari & Keizer, 2021).

The loss of employment and the stress of pregnancy are non-discriminate events impacting individuals across a range of social structures. The distress it will cause to a family varies on a variety of factors, however, no one is entirely immune to these events. The devastation they induce can be amplified by other factors and provoke comorbid situations, such as alcohol abuse or engaging in illicit behaviour. Hence there is a need to address these issues and inquire on accessible protective factors that can be implemented as soon as possible.

Protective features in the case of stress and stress exposure rely heavily on the Bronfenbrenner model and the interactions that each of these systems have with the developing child. Thus, we are not limited to the absence of risk factors and the avoidance of stress, but rather instances that promote well-being and self-security. Recognising the inevitability of stress, there is an incentive to accumulate protective factors to be able to appropriately confront these stressful events (Santee et al., 2023). Observing the potential at the microsystem constraint as acknowledged through stable-parenting and peer support, strengthening the relationship across the mesosystem like the academic setting: beyond grades and exams, but viewed as an opportunity for

learning and growth (Tong & An, 2024). On a more distant level through the exosystem and macrosystem, the availability of third-spaces, economic stability and a sense of community can provide protective factors to strengthen the foundation that is built to deal with stress.

The interventions available for stress have been almost entirely behavioural, with drug interventions coming in the form of prescribed antidepressants such as selective-serotonin reuptake inhibitors. Considering chronic stress' interaction with dopamine, these targeted therapies provide necessary correction and have seen a lot of progress whilst maintaining its scarcity with the requirement of prescription and diagnosis (Nawaz et al., 2024). Stress, like anxiety or depressive disorders, have lifestyle changes as a primary treatment plan, and it is something that has also seen positive outcomes (Sharma et al., 2024).

Consequently, if I were to translate my research into policy recommendations, I would begin with small-scale initiatives within environments already under partial control of policymakers. I would make all tests for high school and preschool students to be exclusively after "break-time" or "lunch-time", with the assumption that this free time is a designated exercise slot of some form. This would take a step in the right direction, without being excessively invasive or inoperable. It is important to note that, as students age and the importance of exams increase, the suggested policy of mandating "break-times" may not be as effective. This would attempt to target stress as a universal entity, working at both parental and personal sources of stress during the early and crucial years of childhood development.

For an intervention aimed directly at the parental source, I would propose a more flexible approach centered on advice and guidelines rather than strict legal mandates, thereby keeping control in the hands of parents. Nevertheless, psychologists have discussed potential measures to minimise the stress that comes with parenting in particular, proposing strategies to increase communication within parental units, encourage dialogue between parents and friends, and to stimulate self-reflection and critical appraisal of parental actions.

Despite extensive research on the impact of parental stress on children's biological and social development, as well as numerous recommendations for mitigating its negative effects, the literature remains limited in its examination of gender-specific aspects of parental roles. The difference a mother's stress can have both during and beyond pregnancy in comparison to a father's stress which requires further investigation. Do maternal figures have more influence on their children's development? In addition I am curious as to the gender difference in processing and dealing with stress, though this would open a whole other can of worms on gender differences in dealing with stressors. In general; however, the published work I have come across has never focused on differentiating between the parental figure, and it is something I hope to continue researching as the importance of both mother and fathers are underlined in healthy child development.

That said, the literature analysis, reflection, and research performed to produce this project has induced a lot of thought on how many risk factors that affect varying levels of the Bronfenbrenner model are known, but perhaps have not been sufficiently considered in policy making. This course has chronologically analysed only a small

quantity of risks that scientists are aware impact child development. However there is still a language barrier between published scientific work and the general public, which delays the benefits that these lengthy studies could potentially provide. While this lag is instituted through state-level policy implementation and overarching institutional regulations, it is felt by smaller, independent organisations such as charities actively fighting child development issues, such as stressors. Ultimately, this project seeks to make the complex relationship between parental stress and child development more understandable by reviewing existing studies and highlighting areas for further research.

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